

Question	Marking Guidance	Mark	Comments
09.1	1. Spiracles, tracheae, tracheoles; 2. Spiracles allow <u>diffusion</u> (of oxygen) OR (Oxygen) <u>diffusion</u> through tracheae/tracheoles; 3. Tracheoles are highly branched so large surface area (for exchange); 4. Tracheole (walls) thin so short diffusion distance (to cells) OR Highly branched tracheoles so short diffusion distance (to cells) OR Tracheoles enter cells so short diffusion distance; 5. Tracheole permeable to oxygen/air; 6. Cuticle/chitin/exoskeleton (impermeable) so reduce water loss; 7. Spiracles (can) close so no/less water loss OR Spiracles have valves so no/less water loss; 8. Hairs around spiracles reduce water loss;	5 max (5 x AO1)	Reference to these 3 structures anywhere in answer = 1mark If whole answer refers to MPs 1–5 only, award MAX 4 marks 3. Accept 'network' or 'large number' for highly branched 3., 4. and 5. If tracheae/tracheoles confused, penalise once only 4. Allow 'next/close to' for enter cells 6. Allow prevents water loss

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09.2	Breathing in 1. Diaphragm (muscles) contract and diaphragm flattens; 2. External intercostal muscles contract and ribcage pulled up/out; 3. (Causes) volume increase and pressure decrease in thoracic cavity (to below atmospheric pressure); Breathing out 4. Diaphragm (muscles) relaxes and diaphragm moves up; 5. External intercostal muscles relax and ribcage moves down/in; 6. (Causes) volume decrease and pressure increase in thoracic cavity (to above atmospheric pressure);	5 max (5 x AO1)	1. Accept 'pulled/moved down' for flattens. 3. and 6. Accept lungs or thorax for 'thoracic cavity' Reject 'chest' once 4. Accept 'returns to domed shape' for moves up 5. Accept internal intercostal muscles contract